Market Power and the Gender Career Gap

Evidence from Matched HR Records in Chile

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Firms shape careers through hiring and promotion

- **The gap:** Gender gap research focuses on wages, but firms also decide who to hire and promote. How does labor market power distort these decisions by gender?
- **At hiring:** Audit studies document callback gaps (Bertrand & Mullainathan, 2004), but cannot observe the full pipeline or link screening to firm characteristics. **Do firms with more market power screen more aggressively against women?**
- **At promotion:** The first promotion is where the pipeline narrows most (Benson et al., 2026), but evidence is single-firm. **If women have fewer outside options, firms can extract rents through slower career progression.**
- **This paper:** tests whether firms with greater labor market power distort career dynamics for women at *both* margins.

The puzzle of Chile: low participation, persistent wage gaps

- Women surpassed men in education, TFR fell to 1.2, yet the wage gap persists (15–20%) and female labor force participation remains low (53% vs. OECD 65%)
- If human capital and fertility can't explain the gap, what can? $\Rightarrow$ **Firm-side frictions:** how firms hire, promote, and retain workers
- Chile is also a **natural laboratory** — overlapping reforms create variation:
  - 40-hour workweek (Ley 21.561) — changes the cost of "greedy jobs"
  - Flexible work for caregivers (Ley 21.645) — shifts women's outside options
  - Childcare mandate at 20 female workers (Art. 203) — RD at a firm-size threshold
- Each reform (ideally) shifts a **parameter of the model**: λ_F (outside options), hours norms, or the cost of female labor

BUK records the full HR pipeline

BUK is an HR platform used by 19,000+ Chilean firms (4.7M workers, 2022–2026). It records each stage of the employment relationship:

Hiring	Contracts	Payroll	Promotions	Separations
Applications, candidates, selection	Job title, area, hours, start date	Base, gross, net salary, bonuses	Title changes, wage changes, hierarchy	End date, exit reason, turnover

- **Worker–month panel** with firm, occupation, and supervisor identifiers
- **Recruitment:** 186K selection processes, 2.7M applications
- **Complemented by** the Work in Progress survey (perceptions, satisfaction, preferences) Survey details

Firms: 19,000+ Chilean firms, skewed toward large employers

	N firms		% of workers
Large (>250 employees)	5,459	(28.4%)	79.8%
Medium (50–250)	6,127	(31.9%)	14.5%
Small (<50)	7,612	(39.6%)	5.7%
Total		**19,358**	**100%**

- Panel covers 2022–2026; firm count grows from 6,508 (2022) to 19,104 (2026)
- 20 SII economic sectors represented; top 5: retail, professional services, admin services, finance, manufacturing
- Linkable to SII tax records for 87% of firms (sector, sales bracket, equity)

Workers: 4.7M employees with detailed job spells and wage records

Unique workers	4,712,667
Job spells	8,273,604
Male / Female	62.8% / 37.2%
Job spells per worker	
Mean	5.86
Median	3.00
P10	1.00
P90	14.00

- **Hierarchy:** supervisor link enables org structure and promotion analysis
- **Wages:** base, gross, and net salary; allows fixed vs. variable decomposition
- **Job characteristics:** area/occupation, contract type, workday type
- **Separations:** end date for 83.2% of completed spells; exit reason coverage low

Recruitment: 186K hiring processes with full application-to-hire pipeline

Selection processes	186,457
Applications	2,725,982
Unique candidates	1,146,914
Median apps per vacancy	2
Mean apps per vacancy	18.7
P90 apps per vacancy	42
Candidate gender: Male / Female	54.5% / 44.5%

- 68.9% of processes closed (hired); 48.3% are replacements, 38.8% new vacancies
- Top referral sources: external portals (40.5%), company portal (23.7%), job boards (14.7%)
- Heavy right tail in selectivity — some vacancies attract 1,000+ applicants

Many possible research questions

1. **The broken rung at first promotion**
   - Is there a gender gap at the first promotion, and what worker and firm characteristics predict it?
   - Do firms and workers sort on availability rather than productivity? Do long-hours norms explain who gets promoted?
2. **Monopsony power and career dynamics**
   - Do firms with greater labor market power offer fewer promotion opportunities to women?
   - Do firms with greater labor market power screen out women or offer lower entry wages?
3. **Firms adjust to mandate cost**
   - How do firms reorganize their workforce in response to mandates? Do they reclassify contract type, substitute towards to male labor, outsource roles?
4. **And many more...**

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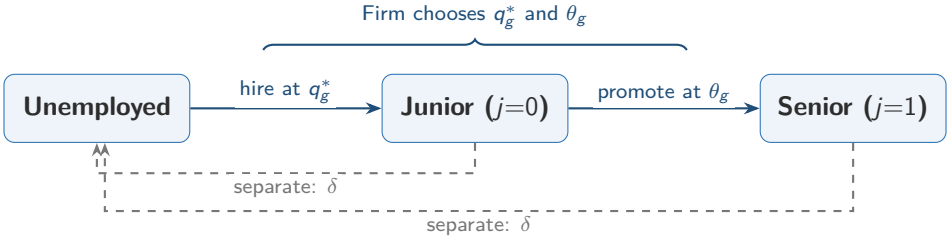
Two literatures have developed in parallel — this paper connects them

- **Monopsony × gender:** market power depresses women's wages (Webber, 2016; Sharma, 2023); outside options causally shape within-firm outcomes (Caldwell & Danieli, 2022) — **but only studied for *wages***
- **Gender × career dynamics:** promotion gaps persist after controlling for performance (Benson et al., 2026); hiring discrimination documented via audits (Bertrand & Mullainathan, 2004) — **but no connection to *market structure***
- **The missing intersection:** Whether market power distorts hiring screening or promotion decisions by gender. Are rents extracted through slower career progression.
- **Contribution:** Connect monopsony to career dynamics at both margins, using application-level data to measure market power directly

Model overview: on-the-job search with hiring and promotion decisions

Building on: Burdett & Mortensen (1998) wage-posting search model, extended with an internal promotion decision (Gibbons & Waldman, 1999) and gender-specific outside options (Sharma, 2023).



- Continuous time, two worker types: $g \in \{M, F\}$
- Firms heterogeneous in productivity $p \sim F(p)$; two job levels per firm
- **Key parameter:** outside offer arrival rate λ_g, with $\lambda_F < \lambda_M$
- Firms choose **two instruments**: screening threshold at hiring and promotion threshold

Workers' problem

Workers of gender g are either **unemployed** or **employed** at level $j \in \{0, 1\}$ in a firm of type p.

Value of unemployment:

$$rU_g = b + \lambda_g^u \int \max\{V_{g0}(w) - U_g,\ 0\}\, dF_g(w)$$

where b is the flow value of unemployment, λ_g^u is the offer arrival rate, and $F_g(w)$ is the wage offer distribution.

Value of employment at level j:

$$\begin{aligned} rV_{gj}(p) =& w_{gj}(p) + \lambda_g \int \max\{V_{g0}(w') - V_{gj}(p),\ 0\}\, dF_g(w') + \delta[U_g - V_{gj}(p)] \\ &+ \mathbb{1}_{j=0} \cdot \mu_g(p)[V_{g1}(p) - V_{g0}(p)] \end{aligned}$$

- λ_g: on-the-job outside offer arrival rate ($\lambda_F < \lambda_M$)
- $\mu_g(p)$: promotion rate at firm p for gender g (determined by firm's choice)

Firm's problem: two instruments

A firm with productivity p employs workers at two levels. Output of gender-g worker at level j:

$$y_j(p) = p \cdot a_j \qquad \text{with } a_1 > a_0$$

The firm chooses **four objects** for each gender g:

Decision 1: Hiring (Shi, 2002)

- Entry wage w_{g0} and screening threshold q_g^*
- Applicants draw quality signal $q \sim H(q)$; hired if $q \geq q_g^*$
- Trade-off: higher $q_g^* \rightarrow$ better workers, fewer hires

Decision 2: Promotion (Gibbons & Waldman, 1999)

- Promotion threshold θ_g and senior wage w_{g1}
- Each period, junior worker draws performance signal $s \sim G(s)$; promoted if $s \geq \theta_g$
- Trade-off: lower $\theta_g \rightarrow$ more promotions (higher output), but higher wage bill
- **Retention motive:** promoting prevents separation to outside offers

Firm's problem: steady-state profit maximization

The firm chooses $(w_{g0}, w_{g1}, q_g^*, \theta_g)$ to maximize steady-state profits:

$$\pi_g(p) = \sum_{j \in \{0,1\}} [y_j(p) - w_{gj}] \cdot n_{gj}(p)$$

where the steady-state workforce $n_{gj}(p)$ depends on three flows:

1. **Inflow** (hiring): $h_g(p) = \lambda_g^u \cdot u_g \cdot [1 - H(q_g^*)] \cdot \mathbb{1}\{V_{g0}(p) \geq U_g\}$
2. **Internal flow** (promotion): $\mu_g(p) = 1 - G(\theta_g)$
3. **Outflow** (separation): $s_g(p) = \delta + \lambda_g \cdot [1 - F_g(V_{gj}(p))]$

In steady state, the mass at each level satisfies:

$$\dot{n}_{g0} = h_g(p) - [\mu_g(p) + s_g(p)] \cdot n_{g0} = 0$$
$$\dot{n}_{g1} = \mu_g(p) \cdot n_{g0} - s_g(p) \cdot n_{g1} = 0$$

The retention trade-off: why monopsony distorts promotions

The firm's promotion decision balances **two forces**:

1. **Direct profit effect:** promoting changes per-worker surplus
   Output rises ($a_1 > a_0$), but so does the wage ($w_{g1} > w_{g0}$).

2. **Retention effect:** promoting makes the worker less likely to leave. Promotion $\rightarrow$ $V_{1g} > V_{0g}$ $\rightarrow$ fewer outside offers are attractive $\rightarrow$ lower separation.

Men (high λ_M):

- "If I don't promote him, he'll leave"
- Retention urgency is **high**

$\Rightarrow$ Promote to retain

Women (low λ_F):

- "Even without promotion, she won't leave"
- Retention urgency is **low**

$\Rightarrow$ Delay is low-cost

$\Rightarrow$ $\theta_F^* > \theta_M^*$: higher bar for women, not from productivity differences, but because firms face **less competitive pressure** to promote them.

Formal FOC

The hiring mechanism: why monopsony distorts screening

When $\lambda_F < \lambda_M$, two **opposing forces** shape the firm's screening threshold q_g^*:

1. **"Longer tenure" force** $\rightarrow$ screen women *less*
   Women stay longer at the firm $\rightarrow$ each hire generates surplus over more periods $\rightarrow$ the expected value of a marginal hire is higher $\rightarrow$ firm should lower the bar to hire more women

2. **"Less competition" force** $\rightarrow$ screen women *more*
   Women have fewer outside options $\rightarrow$ they apply even when the bar is high $\rightarrow$ firm can raise q_F^* without losing its applicant pool $\rightarrow$ "cherry-pick"

The sign of $q_F^* - q_M^*$ is ambiguous

Test: BUK application-to-hire conversion rates by gender, across markets with different concentration, reveal which force dominates in practice.

Counterfactual exercises

With the estimated model, compute:

1. **Equalize outside options:** set $\lambda_F = \lambda_M$
   $\rightarrow$ What fraction of the gender promotion gap is explained by differential outside options vs. other forces (taste discrimination, statistical discrimination)?

2. **Eliminate market power:** set all markets to perfectly competitive ($\lambda \rightarrow \infty$)
   $\rightarrow$ How much of the gender gap in career dynamics would disappear under perfect competition?

3. **Policy simulation — 40-hour workweek:** if the reform increases λ_F (more firms become viable options for women), how much does the promotion gap close?
   $\rightarrow$ Connects the structural model to Chile's ongoing labor market reforms

Work in Progress 2026 Survey

- **Sample:** 6,789 respondents in Chile, Colombia, Mexico, Peru
- **Open survey** — anyone can respond; **cannot be linked** to platform data
- **Topics covered:**
  - Demographics, salary, job tenure, firm size
  - AI adoption at work, training, innovation
  - Job satisfaction, burnout, work-life balance
  - Gender equity perceptions, discrimination, D&I policies
  - Benefits, performance evaluation, retention intentions
- Provides **subjective** complement to BUK's **administrative** records

Source: BUK Work in Progress Survey 2026.

Back

Formal promotion condition (two-period simplification)

Setup: Period 1, worker is junior (profit $\pi_0 = pa_0 - w_{g0}$). Between periods, outside offer arrives with probability λ_g. If promoted, continuation value rises: $V_{1g} > V_{0g}$.

Retention probabilities:

$$r_g^P = 1 - \lambda_g[1 - F_g(V_{1g})] \quad \text{(if promoted)}$$

$$r_g^N = 1 - \lambda_g[1 - F_g(V_{0g})] \quad \text{(if not promoted)}$$

Since $V_{1g} > V_{0g}$: $r_g^P > r_g^N$ (promoted workers are easier to retain).

Promote if period-2 expected profit is higher with promotion:

$$\underbrace{(\pi_1 - \pi_0) \cdot r_g^N}_{\text{Direct profit effect}} + \underbrace{\pi_1 \cdot (r_g^P - r_g^N)}_{\substack{\text{Retention value} \\ \text{of promoting}}} \geq 0$$

The retention term is proportional to λ_g: $\quad r_g^P - r_g^N = \lambda_g \left[F_g(V_{1g}) - F_g(V_{0g})\right]$

- As $\lambda_g \to 0$: retention term vanishes $\to$ promote only if $\pi_1 > \pi_0$
- As $\lambda_g \to \infty$: retention dominates $\to$ always promote to retain
- Since $\lambda_F < \lambda_M$: retention motive is weaker for women $\to \theta_F^* > \theta_M^*$